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## **Section A : Validation Criteria and Checklist**

### **Q1. Validation Criteria Engineering**

#### **1. Measurable Validation Criteria**

| System Feature               | Stakeholder Expectation        | Validation Criteria (Measurable)                                                           |
|------------------------------|--------------------------------|--------------------------------------------------------------------------------------------|
| Student Login                | Secure and reliable access     | invalid login attempts blocked after 5 tries and response time < 3 seconds                 |
| Online Assignment Submission | Correct and timely submission  | File upload success rate should be $\geq 98\%$ and accepted file formats are validated     |
| Attendance Tracking          | Accurate attendance records    | attendance reports should be generated within 10 seconds.                                  |
| Faculty Dashboard            | Easy monitoring of students    | Dashboard should load within 4 seconds and it displays correct student and assignment data |
| Activity Logging             | Traceability of system actions | Every login, upload, and update event logged with correct date and time.                   |

#### **2. Structured Validation Checklist**

| ID   | Module                | Validation Item           | Expected Result                | Test Method           | Status | Evidence        |
|------|-----------------------|---------------------------|--------------------------------|-----------------------|--------|-----------------|
| VC 1 | Student Login         | Valid credentials login   | User redirected to dashboard   | Functional Test       | Pass   | Screenshot/ Log |
| VC 2 | Student Login         | Invalid password handling | Error message displayed        | Negative Test         | Pass   | Screenshot      |
| VC 3 | Assignment Submission | Upload PDF assignment     | File uploaded successfully     | Functional Test       | Pass   | Upload Log      |
| VC 4 | Attendance Tracking   | Mark attendance           | Attendance stored correctly    | Database Verification | Pass   | DB Record       |
| VC 5 | Faculty Dashboard     | Dashboard loading         | Loads within 4 sec             | Performance Test      | Pass   | Performance Log |
| VC 6 | Activity Logging      | User activity recording   | Logs generated with timestamps | Log Verification      | Pass   | Log File        |

## **Q2. Validation Automation using GitHub Actions**

### **1. Role of CI Pipelines in Continuous Validation**

Continuous Integration (CI) pipelines automate software validation whenever developers commit code changes. Their role includes:

- Automatically building the application
- Running automated tests
- Detecting bugs early
- Generating validation evidence and logs
- Ensuring software quality before deployment
- Supporting continuous monitoring and compliance

Benefits include faster feedback, reduced manual testing effort, and improved software reliability.

## **Section C : Security Validation and Vulnerability Assessment**

### **Q3. Structural Vulnerability Examination using OWASP ZAP**

#### **1. Importance of Security-Oriented Validation**

Security-oriented validation ensures that software systems are protected against cyber threats and vulnerabilities before deployment. It helps maintain confidentiality, integrity, and availability of data while reducing the risk of attacks, data breaches, and financial loss.

#### **2. Three Vulnerabilities**

##### **a) SQL Injection**

Occurs when attackers insert malicious SQL queries into input fields to manipulate databases.

**Impact:** Unauthorized data access, deletion, or modification.

##### **b) Cross-Site Scripting (XSS)**

Occurs when malicious scripts are injected into web pages viewed by users.

**Impact:** Session hijacking, cookie theft, and malicious redirects.

##### **c) Broken Authentication**

Occurs when authentication mechanisms are poorly implemented.

**Impact:** Attackers may gain unauthorized access to user accounts.

**Section D : Validation Risk and Compliance Documentation**

**Q4. Validation Risk Management and Compliance Evidence**

**1. Validation Risk Register**

| <b>Risk ID</b> | <b>Risk Description</b>                  | <b>Impact</b> | <b>Probability</b> | <b>Mitigation Strategy</b>                |
|----------------|------------------------------------------|---------------|--------------------|-------------------------------------------|
| VR-01          | Unauthorized access to financial records | High          | Medium             | Implement MFA and RBAC                    |
| VR-02          | Sensitive customer data leakage          | High          | Medium             | Encrypt data in transit and at rest       |
| VR-03          | Missing audit logs                       | Medium        | Low                | Enable centralized logging and monitoring |